ENGR 120/121 – Design Project Planning

192.168.4.1

Lab section: B09

Lab TA Name: Amir Mehdirezaei Khamse

Team ID: 02

Team member's V-number	Team member's last name	Team member's first name
V01089158	Boulanger	Justin
V01054513	Wai	Min
V01064432	Clark	Dillon

Team name (optional): UVIC Security Systems Reinforcement

Team logo and caption (optional): Everyone deserves security

Problem definition

1) What is the problem your team is solving?

   The problem our team wishes to address is the security on the UVIC campus during later hours of the night. The university has suffered from multiple B&Es, thefts and intrusions. These crimes occur every year, impacting many students' lives and safety [1].

   The university of Victoria suffers from a multitude of B&Es and intrusions causing thefts among students as well as the consistent need for building checks by security.

2) How did your team select the problem?

- Defined a few common issues UVIC may face with systems provided to us, such as:
  - Effective cooling in specific areas
  - Ambient lighting in study areas
  - Limited visibility in construction area at night
- Deliberated on key factors of each problem using a chart, including:
  - Severity of issue
  - Feasibility based on our resources
  - Physical aspects of issue
  - People affected by issue
  - Consistency of issue
- Reviewed existing protocols within UVIC Security team
- Assessed how to seamlessly incorporate multiple systems to achieve the desired grade

3) Why did your team choose this problem?

Based on the factors listed above, we concluded that this problem has the highest prevalence while also remaining attainable within the bounds of our resources. UVIC remains a victim of crime due to the multitude of individuals constantly entering and exiting the campus throughout various hours of the day. Our system could relieve the work of patrolling security by deterring unlawful activities in low light scenarios.

Proposed Solution

4) What is your team's proposed solution to the problem?

Our proposed solution incorporates a reed switch (magnetic sensor) that checks whether a door is opened or closed during closing hours of the day, and a response that alerts security of suspicious activity. When the magnetic field of the reed switch breaks, our code will send an alert to security and light up the area indicated by the sensor. An RFID scanner will be used to check if a person has permission to enter the room.

5) How did your team arrive at this solution?

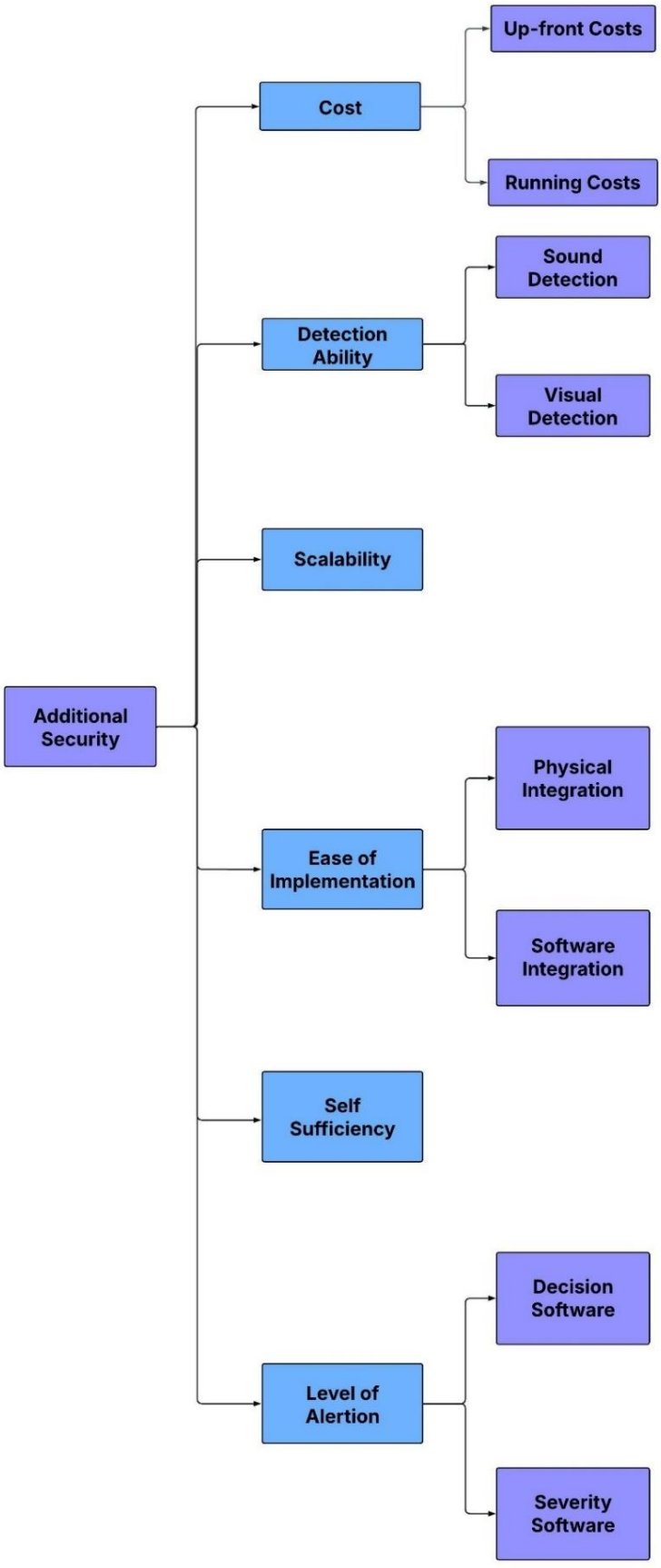
Initial brainstorming occurred as a team and notes jotted down of the types of systems possible. We created a weighted criteria tree to brainstorm general solutions for individual problems within the overarching issue. After deliberation within the team and utilizing our created criteria tree, we came to the decision of pursuing this general solution as it best fit our idealized tree.

Additionally, a study by J. M. Macdonald *et al.* [2] examines the recent addition of better lighting in urban areas in Philadelphia. The study takes various qualitative and quantitative measurements which result in a conclusion that nighttime street crimes declined by 15% [1]. Although Macdonald's study gathers from a much larger area of effect, the principle applies to helping at UVIC.

Finally, our team crafted a morphological chart with a variety of solutions in aspects:

-----	1	2	3	4	5
Sense	Movement Sensor	Mechanical Switch	Light Sensor	Magnetic sensor	Pressure Sensor
Bypass	OneCard	Keypad	Button	--------	--------
Alert	Noise Alarm	Visual Alarm	Notification	--------	--------
Illumination	LED	Floodlight	Flashlight	Beam Light	--------

Concepts Integrated	One concept	Two Concepts	Three concepts	Four concepts	--------

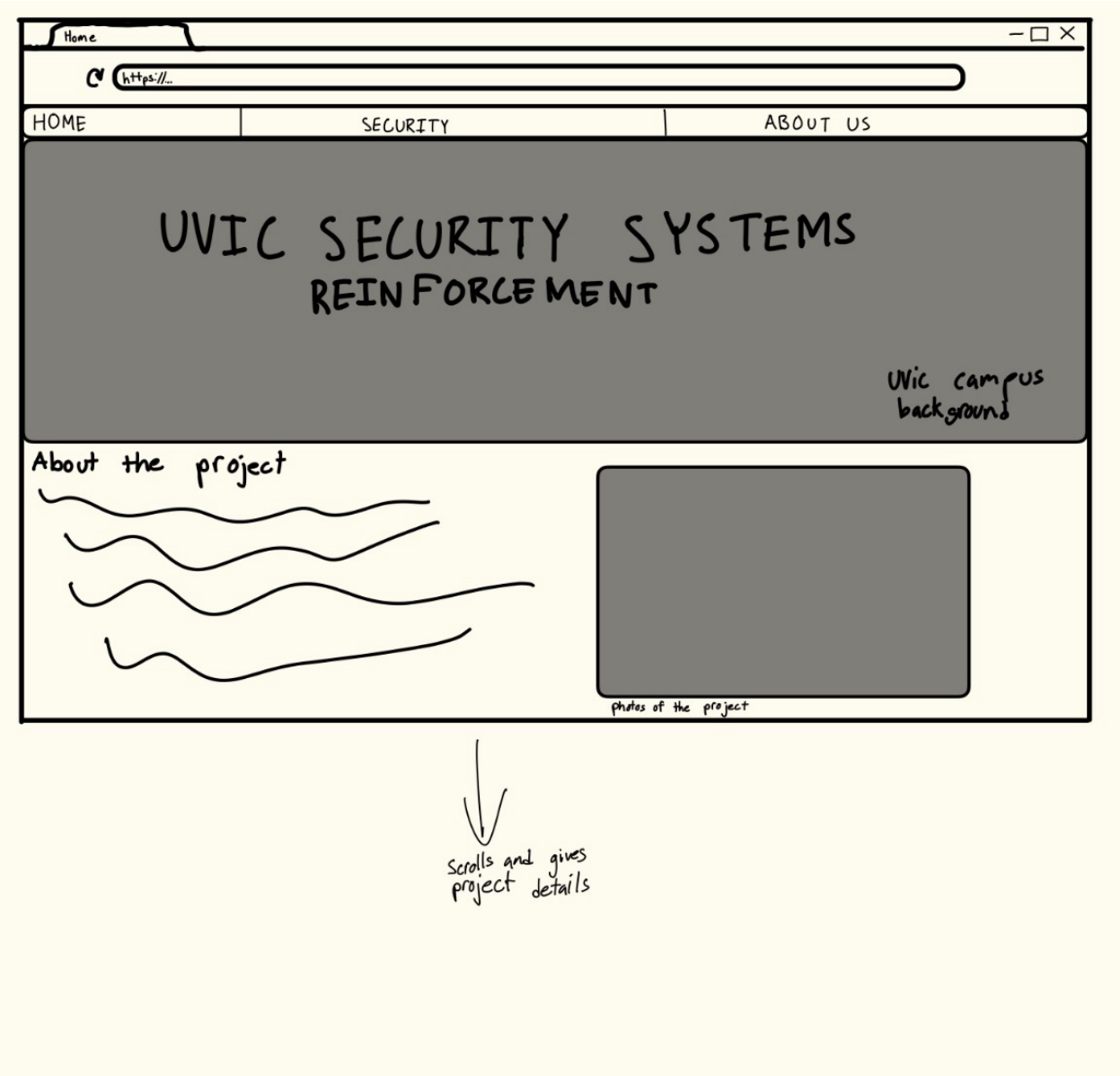

Additional Security
Cost
Up-front Costs
Running Costs
Detection Ability
Sound Detection
Visual Detection
Scalability
Ease of Implementation
Physical Integration
Software Integration
Self Sufficiency
Level of Alertion
Decision Software
Severity Software

6) Technical details of your team's proposed solution

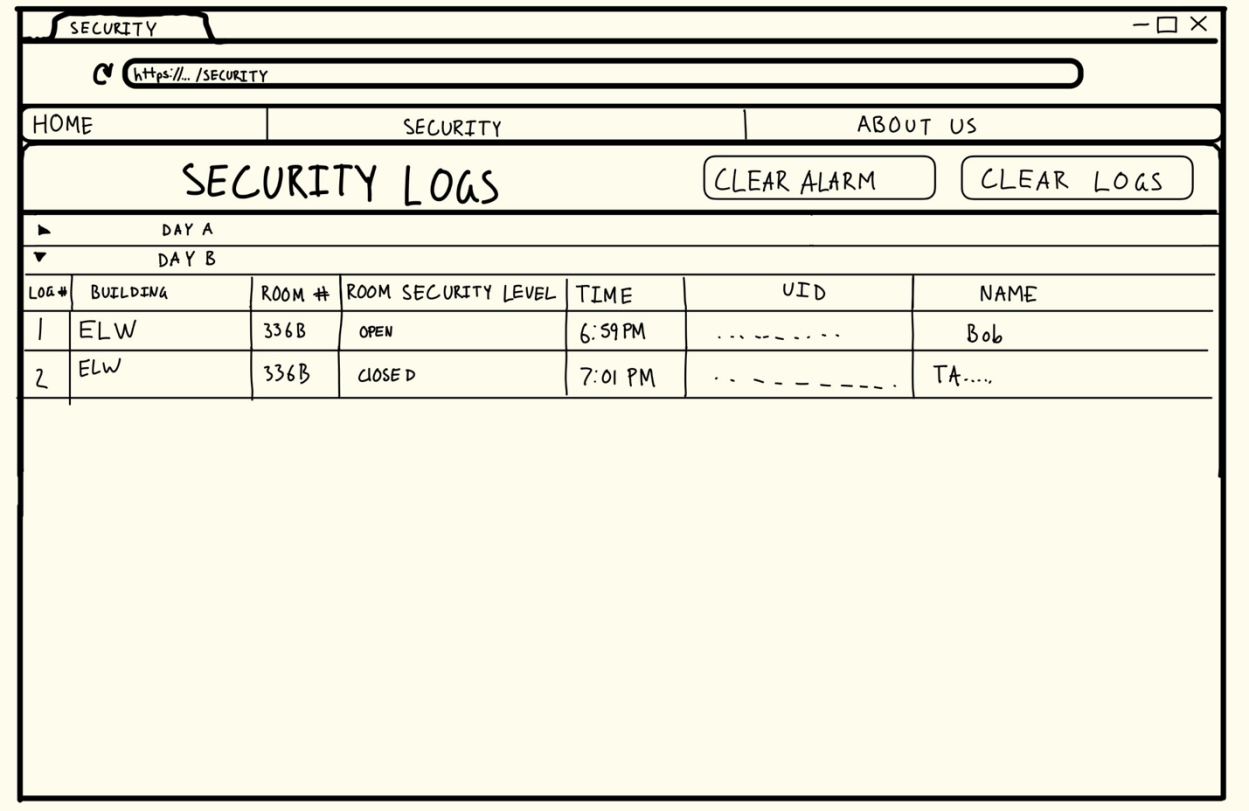
6a) List all the sensors and actuators that would be used. Justify their need.

Component	Quantity	Need
Reed Switch	2	Detect magnetic field for open/closing of door
RFID Sensor (RC522)	1	Scan for security bypass
LED	2	Light when circuit is broken/door is opened
Buzzer	1	Audible alert for security when door is opened
PIR	1	Detect motion in area
Resistors	Several	Circuit protection

6b) Supply a sketch of the user-interface you would be developing.



Home page



Security log page. Some of the data may change, but this is generally what it will look like.

7) Roles & responsibilities:
Each team member is expected to contribute towards the project. Circuit designing, Pico programming and User interface design using HTML coding are the three different aspects of the project. Please use the table below and indicate each team member's responsibilities.

Team Member	Circuit designing	Pico programming	HTML coding
Justin Boulanger (Team member 1)	Primary circuit designer • Procurement of sensors • RFID circuit design • Integration of all systems on circuit	Secondary Pico Programmer • Develop whitelist of RFID scans • Security override code	Tertiary HTML Programmer • Override button • List of persons scanning
Dillon Clark (Team member 2)	Secondary Circuit Designer • Assistant circuit designer • Motion detection for lighting system	Tertiary Pico Programmer • Merging html and MicroPython code • Coding the motion detection system.	Primary HTML Designer and Programmer -Coding different pages (about us, security alerts, and our project description) -Building the primary UI and functionality of the http route navigation (/, /about, /home etc)

Min Wai (Team member 3)	Tertiary Circuit Designer • Reed switch design	Primary Pico Programmer • Develop the main micro-python control logic • Implement the magnetic door sensor and motion sensor control system • Program the LED and Buzzer activation when intrusion is detected.	Secondary HTML Programmer • Aids in Webpage coding • Updating the webpage • Displaying the real-time security system status

8) Project planning:

8a) Each team member must list the activities they would be doing and the timeline by which they would be delivering their tasks.

Team member 1 (Justin):

Task	How will you test successful completion?	Date of delivery
Simple buzzer circuit to create a noise	Power supplied, the buzzer will create a noise.	March 5th
Add RFID bypass	Circuit will be tested by providing power to buzzer which then can be stopped with a RFID scan.	March 8th
Develop a whitelist in the python code with user IDs	Same circuit test as previous but override with RFID should not work with certain cards/fobs	March 10th
UI HTML design of visible whitelist and override button	Circuit will remain the same but can be turned off with a button on HTML website as well as RFID scan. Scanning will show who, when and which “door” was accessed/denied	March 15-20th
Utilization of Reed Switch in circuit and	Reed switch sends correct voltage signal to Pico W readable with Thonny or multimeter	March 18th

physical connection with existing circuit		
Addition of PIR lighting system and seamless integration between areas	Circuit will initially be tested by ensure power is supplied correctly with enough voltage to each aspect of the project such that they work independently in the same space	March 20th
Clean up of circuit area and creation of a visible simulation space	Circuit still works and is visually pleasing for demonstration	March 22nd
Run over of entire system in all areas with team, final troubleshooting, testing, etc.	Circuit works as intended and is ready for presentation	March 25th (Day of presentation)

Team member 2 (Dillon):

Task	How will you test successful completion?	Date of delivery
Make objects for both fobs and rooms	Fobs store: UID, name, security clearance, position, expiration date Rooms store: building name, room number, Open hours start, open hours end, normally accepted Fob UIDs, closed accepted fob UIDs	March 3rd
Generate a hashmap for storing fobs	When given a UID all the fob's information can be displayed	March 3rd
HTML website navigation including an about page, a home page (how it works), and a security page.	Can move between /, /about, and /security pages and has functional buttons to switch between pages	March 6th
HTML manual security updates	HTML updates security messages from the MicroPython when the webpage is refreshed	March 11th

HTML clear security feed button for testing	When pressed, the security logs are cleared (for demonstration purposes)	March 11th
Set-up HTML with event driven updates	When a security ping triggers it automatically gets written to the webpage without the need to refresh.	15th March
HTML UI design	Team agrees that colours and formatting of the document is presentable and looks good	20th March
HTML graphic design (pictures)	Team agrees that photos look good and the website is officially complete	25th March

Team member 3 (Min):

Task	How will you test successful completion?	Date of delivery
Main MicroPython control logic	Test by simulating door open/close conditions to verify correct system response.	March 12th
Implementation of sensor logic	Test if the program can display the correct door status (open/close) by moving the magnet toward and away from the reed switch and if the program can detect using the motion sensor when there is a motion	March 18th
Programming LED and buzzer alert	Test if LEDs turn on and buzzer activate immediately once reed switch is opened without using the bypass method or the sensor logic detects a motion in the intrusion condition. Test if only LEDs turn on and the program does not activate buzzer once the sensor logic detects a motion while using the bypass method	March 19th
Support integration between MicroPython output and webpage content	Once the MicroPython program updates the system state from the reed switch input, verify that the correct values are transmitted to the webpage and displayed without missing or incorrect information.	March 21st
Updating system status display in webpage	Once the MicroPython program detects a change in the reed switch input, verify that the webpage updates to display the correct system status (e.g., NORMAL vs ALERT) and sensor state.	March 22nd

UI feedback improvement	Test all buttons, status indicators, and alert messages to ensure they respond correctly and clearly communicate the system's current state to the user.	March 23rd

8b) What is your team's strategy if one of the members is unable to complete their tasks on time?

In the event one of our members is unable to complete a task on time, work will be quickly redistributed among available members. The covering team members will work with the original member to organize the new timeline of completion. If needed, the original member who was unable to complete their task may be offered other tasks which better suit their skillset. The team member who failed to complete their task may be reported to the TA/Prof. if it is determined that their failure to complete work is due to a lack of effort or negligence. In the event of major project failures, we will organize a meeting with the instructors to gain insight on an actionable path in a timely matter. Individual unfortunate circumstances will be easily accommodated and worked around as we all have differing student schedules.

8c) What challenges do you foresee in the successful integration of team members' work? How do you plan to mitigate them?

- Continuous swapping/exchanging of materials when needed - everyone may need the breadboard at a certain time to do different pieces of the project.
    - Ideal solution is to organize schedules, so we are working on different aspects of HTML/Circuit/Python with no overlaps
    - If things do overlap, there may be a need for minor changes to scheduling.
    - Focus can be shifted and some tasks can be swapped
- Combination of each circuit and code – each team member has their own style and idealistic coding/designing systems.
    - Effective communication during each step and defining markers within code can prevent this.
    - Ensuring that each team member focuses on their individual circuit pieces may help speed up work and allow members to better understand each part.
- Overlapping of tasks – with each team member working on different areas, there may be times when a team member must wait on another to complete a prerequisite task
    - Similar solutions to the exchanging of materials problem, tasks can be rearranged or the waiting team member can assist in accelerating the necessary task

    - Meeting regularly to update each other on progress
    - Keeping backup tasks so members can continue working on while waiting on another part.

- Communication delays – differences in availability or response time may slow project progress.
    - Use a group chat or shared document to keep updates visible to all members.
    - Clearly document any changes made to code or circuit so others can adjust.

References

[1] "Crime statistics - University of Victoria," *Uvic.ca*, 2012. https://www.uvic.ca/security/home/prevention/statistics/index.php

[2] J. M. Macdonald *et al.*, "Can Enhanced Street Lighting Improve Public Safety at Scale? 1," Jan. 2025, doi: https://doi.org/10.2139/ssrn.5150459.

[3] P. Delgoshaei, M. Heidarinejad, K. Xu, J. R. Wentz, P. Delgoshaei, and J. Srebric, "Impacts of building operational schedules and occupants on the lighting energy consumption patterns of an office space," *Building simulation*, vol. 10, no. 4, pp. 447–458, 2017, doi: 10.1007/s12273-016-0345-9. Accessed: Feb. 26, 2026.